# Answer key Tutors only

The numbers the web pages show. ADC in × $10^{-3}$ mm²/s. "Say" = the one point to make at the debrief.

## A Same brain, different ADC

ADC & IVIM sandbox

### TAB 1 · SAME BRAIN, DIFFERENT ADC

1 **CSF (ventricles, cisterns) and the vessels in the Sylvian fissure: about a third of their signal goes between b 0 and b 100. Brain tissue loses about 9 %.**
Say: At low b-values, moving blood and CSF dominate the signal. That is not the diffusion you want to measure.

2 **Brain tissue: b 0 & 800 `0.79` · b 200 & 800 `0.75`**

3 **Vessels + CSF (Sylvian fissure): `+69 %` · brain `+5.2 %` · CSF `+5.8 %`**
Say: More blood flow in the region means a bigger jump when b 0 is in the fit. Most pairs guess CSF.

4 **One scan with b 0, one without: 5 % of the 11 %, so about half the limit is used up before anything has changed. Both with b 0: the offset is in both and mostly cancels, but only if the blood flow stays the same (see Tab 2).**

### TAB 2 · TREATMENT (SIMULATED: F 0.25 → 0.08; D = 1.0 AND D* = 20 DO NOT CHANGE)

1 **Guess: with b 0 the ADC falls; with b 200 it stays the same. Real values:**

| | Before | After | Change |
|---|---|---|---|
| b 0 & 800 | 1.36 | 1.10 | **−19 %** |
| b 200 & 800 | 1.01 | 1.00 | −0.9 % |
| b 50 & 800 (task 4) | 1.16 | 1.04 | −10 % |

2–3 **Depends on the b-values. With b 0, ADC falls 19 %: above the 11 % limit, so it counts as a real change. A falling ADC reads as no response. With b 200 nothing changes.**
Say: The cells never changed, only the blood did. The b-values decided the verdict.

4 **No. b 50 halves the effect (−10 %) but does not remove it, and −10 % is just under the 11 % limit. The Unity MR-linac consortium advises a lowest b of 100-150 (Kooreman 2020).**

### TAB 3 · NOISE

1–2 **Clear: D. Noise: f and D* (D* is the worst).**

3 **f is high, usually 0.5-1.0 (median 0.9), different in each voxel. It makes no sense: CSF has no capillary blood.**
Say: CSF's fast free diffusion looks like a "perfusion" component to the model. A fitted number is not proof that the model applies.

4–5 **SNR 50: D stays close (about 0.9-1.1, true 1.0). f wanders (0.09-0.20, true 0.15). D* ranges from 13 to 40 (true 20).**
**SNR 20: only D, and even D spreads ±30 %. f runs from 0 to 0.27, D* from 4 to 150.**
Say: This is why IVIM maps of one voxel are hard to trust, and why published f values disagree.

Ranges = the middle 80 % of 2,000 simulated scans. Ten scans give a similar but noisier picture.

Brain data: E. Peterson, IVIM dataset, Figshare 2016, doi:10.6084/m9.figshare.3395704 (CC0). Limits: QIBA DWI Profile, Stage 3, 2022 (brain 11 %, prostate 47 %). Lowest b-value: Kooreman et al., Radiother Oncol 2020;153:106–13.

## B Has the tumour responded?

Patient PCAMPMRI-00011, shown in reverse order

2 **Scan 1 `1.192` · Scan 2 `1.310` · Change `+10 %`** (volume 0.55 → 0.39 cm³, −28 %)

3

| ROI on scan 1 | Mean | Median | 10th percentile | Change, scan 1 → 2 (mean) |
|---|---|---|---|---|
| Expert, on ADC | 1.192 | 1.187 | 0.984 | +10 % |
| Grown 2 mm | 1.283 | 1.248 | 1.011 | +4 % |
| Shrunk 2 mm | 1.119 | 1.142 | 0.983 | +14 % |
| Expert, on T2w | 1.503 | 1.560 | 1.192 | **−11 %** |

**Moves least: the 10th percentile. It stays at 0.98-1.01 on the three ADC outlines, while the mean runs from 1.12 to 1.28. The T2w outline shifts every value, because on the ADC map it covers partly different tissue.**

Say: The ROI alone moves the mean by −6 % to +26 % on one scan. Low percentiles are less sensitive to the edge.

4 **Cannot tell. +10 % is far below the prostate tumour limit (42 %, Fedorov 2017; 47 %, QIBA). −28 % volume is well within the volume limit (112 % on ADC).**

> **Reveal:** there was no treatment. The two scans are a test-retest pair 13 days apart, shown in reverse. In the order they were taken, ADC *fell* 9 %. Across the 10 men in the study, ADC changed between −9 % and +43 % with no treatment.

5 **Yes. Outline drawn on ADC: +10 %. The same radiologist's outline drawn on T2w: −11 %. Opposite directions.**

## C Contouring on DWI

Patient PCAMPMRI-00007, first scan · 3 mm slices

1–3 **Pairs' own numbers. For comparison, the radiologist: 0.38 cm³ on T2w, 0.37 cm³ on ADC. The ruler should show a few mm at the back of the prostate (the radiologist's two outlines: 2.8 mm front-back).**

Say: Similar volumes do not mean the same place. Look at the overlap and the shift, not only the size.

Bonus **Behind the blue line, towards the balloon: about 5 mm on the middle slices, up to about 10 mm in places. The front edge stays close to the line.**

Say: The whole prostate is pushed towards the air, not only the tumour. An outline drawn on DWI lands several mm too far back on the planning images.

4 **Radiologist: Dice `0.23` · DWI outline `2.8 mm` further back (`5.1 mm` between the centres in all, partly towards the feet).**

Say: If you can't find the T2w outline, check the slice: T2w is drawn on slices 8–11, ADC on slices 7–9.

### DISCUSS (NO SINGLE RIGHT ANSWER)

1 **Most centres: a guide, or a boost volume after checking the geometry. In FLAME the boost was contoured on multiparametric MRI (T2w, DWI and DCE together), not on DWI alone.**

2 **At least the shift measured here: a few mm (5 mm between the radiologist's outlines). It changes from patient to patient, so it can't be a fixed number.**

3 **Rigid registration does not fix it: the distortion is local and non-linear. Distortion correction helps (reverse phase-encoding scan or a B0 field map). Check the prostate edge on b 0 against T2w, especially near the rectal air or balloon.**

### Checklist · suggested answers

| | |
|---|---|
| **1 · Which b-values?** | Lowest b at least 100–150 (not b 0), and the same b-values on every scan. |
| **2 · Whose ROI?** | The same reader and the same method each time, drawn on the ADC map. Report the median or a low percentile too. |
| **3 · Bigger than the repeatability?** | Brain 11 %, prostate tumour 42–47 %, measured on the same scanner with the same protocol. |
| **4 · Geometrically trustworthy?** | Compare DWI with T2w at the edges, near air. Correct the distortion or add a margin. |

Prostate data: QIN-PROSTATE-Repeatability, TCIA, doi:10.7937/K9/TCIA.2018.MR1CKGND (CC BY 4.0); Fedorov et al., Sci Data 2018;5:180281 and Invest Radiol 2017;52:538-46. FLAME: Kerkmeijer et al., J Clin Oncol 2021;39:787-96.